

SAFETY DATA SHEET



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1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product name : SANCTUM®

Manufacturer or supplier's details

Company : FMC Agricultural Solutions A/S

Address : Thyborønvej 78
DK-7673 Harbøre
Denmark

Telephone : +45 9690 9690

Emergency telephone : +44 20 3885 0382 (CHEMTREC's European Regional Toll-Free Number)
1 703 / 741-5970 (CHEMTREC - International)
1 703 / 527-3887 (CHEMTREC - Alternate)

Medical Emergency Number : All other countries: +1 651 / 632-6793 (Collect)

E-mail address : SDS-Info@fmc.com

Recommended use of the chemical and restrictions on use

Recommended use : Insecticide

Restrictions on use : Use as recommended by the label.

2. HAZARDS IDENTIFICATION

GHS Classification

Acute toxicity (Oral) : Category 3

Acute toxicity (Inhalation) : Category 4

Acute toxicity (Dermal) : Category 4

Skin irritation : Category 2

Serious eye damage : Category 1

Skin sensitization : Category 1

Specific target organ toxicity - : Category 2 (Nervous system)
repeated exposure

Specific target organ toxicity - : Category 2 (Respiratory system)
repeated exposure
(Inhalation)

Aspiration hazard : Category 1

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Short-term (acute) aquatic hazard : Category 1

Long-term (chronic) aquatic hazard : Category 1

GHS-Labeling

Hazard pictograms :



Signal Word : DANGER

Hazard Statements : H301 Toxic if swallowed.
H304 May be fatal if swallowed and enters airways.
H312 + H332 Harmful in contact with skin or if inhaled.
H315 Causes skin irritation.
H317 May cause an allergic skin reaction.
H318 Causes serious eye damage.
H373 May cause damage to organs (Nervous system) through prolonged or repeated exposure.
H373 May cause damage to organs (Respiratory system) through prolonged or repeated exposure if inhaled.
H410 Very toxic to aquatic life with long lasting effects.

Precautionary Statements : **Prevention:**
P260 Do not breathe mist or vapors.
P273 Avoid release to the environment.
P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.
Response:
P301 + P310 + P330 IF SWALLOWED: Immediately call a POISON CENTER/ doctor. Rinse mouth.
P305 + P351 + P338 + P310 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/ doctor.
P331 Do NOT induce vomiting.
P391 Collect spillage.

Other hazards which do not result in classification

None known.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Pure substance/mixture : Mixture

Components

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Chemical name	CAS-No.	Classification	MAC value mg/m ³ / TSEL value	Concentration (% w/w)
malathion (ISO) [containing ≤ 0,03 % isomalathion]	121-75-5	Acute Tox.4; H302 Acute Tox.5; H333 Acute Tox.5; H313 Skin Sens.1; H317 Aquatic Acute1; H400 Aquatic Chronic1; H410	No data available	≥ 50 - < 70
Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified	64742-94-5	Asp. Tox.1; H304 Aquatic Acute2; H401	No data available	≥ 10 - < 20
Poly(oxy-1,2-ethanediyl), α-hydro-ω-hydroxy-, mono-C8-10-alkyl ethers, phosphates	68130-47-2	Skin Corr.1B; H314 Eye Dam.1; H318 Aquatic Acute2; H401	No data available	≥ 5 - < 10
maleic anhydride	108-31-6	Acute Tox.4; H302 Acute Tox.5; H313 Skin Corr.1B; H314 Eye Dam.1; H318 Skin Sens.1; H317 STOT RE1; H372 (Respiratory system) Aquatic Acute3; H402	MPC-STEL: 1 mg/m ³ Class 2 - Highly dangerous, Allergens, Substances which require special skin and eye protection Data Source: RU OEL	≥ 3 - < 5
GAMMA-CYHALOTHRIN	76703-62-3	Acute Tox.3; H301 Acute Tox.1; H330 Acute Tox.4; H312 Skin Irrit.2; H315 Eye Irrit.2A; H319 Skin Sens.1; H317 STOT RE1;	No data available	≥ 1 - < 2.5

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		H372 (Nervous system) Aquatic Acute1; H400 Aquatic Chronic1; H410		
phosphoric acid	7664-38-2	Met. Corr.1; H290 Acute Tox.5; H303 Skin Corr.1A; H314 Eye Dam.1; H318 Aquatic Acute2; H401 Aquatic Chronic2; H411	TSEL: 1 mg/m3 Data Source: RU TSEL	>= 0.25 - < 1

For explanation of abbreviations see section 16.

4. FIRST AID MEASURES

- General advice : Move out of dangerous area.
Consult a physician.
Show this material safety data sheet to the doctor in attendance.
Symptoms of poisoning may appear several hours later.
Do not leave the victim unattended.
- If inhaled : Remove to fresh air.
If unconscious, place in recovery position and seek medical advice.
If experiencing any discomfort, immediately remove from exposure. Light cases: Keep person under surveillance. Get medical attention immediately if symptoms develop. Serious cases: Get medical attention immediately or call for an ambulance.
- In case of skin contact : If on clothes, remove clothes.
If on skin, rinse well with water.
Wash off immediately with soap and plenty of water.
Get medical attention immediately if irritation develops and persists.
Wash contaminated clothing before re-use.
- In case of eye contact : Small amounts splashed into eyes can cause irreversible tissue damage and blindness.
In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
Continue rinsing eyes during transport to hospital.
Remove contact lenses.

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- Protect unharmed eye.
Keep eye wide open while rinsing.
Take victim immediately to hospital.
- If swallowed : Keep respiratory tract clear.
Do NOT induce vomiting.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
Take victim immediately to hospital.
- Most important symptoms and effects, both acute and delayed : Gamma-cyhalothrin can cause feelings of burning, tingling or numbness in exposed areas (paraesthesia).
Exposure may result in nausea, vomiting, tremors, cramps, weakness, shortness of breath, a slowed heart rate, headache, abdominal pain, and diarrhea.
On exposure to larger quantities of aged product, symptoms of poisoning (cholinesterase inhibition) may occur.
Toxic if swallowed.
May be fatal if swallowed and enters airways.
Harmful in contact with skin or if inhaled.
Causes skin irritation.
May cause an allergic skin reaction.
Causes serious eye damage.
May cause damage to organs through prolonged or repeated exposure.
- Protection of first-aiders : Avoid inhalation, ingestion and contact with skin and eyes.
- Notes to physician : If there is any sign of poisoning, call a doctor (physician), clinic or hospital immediately. Explain that the victim has been exposed to a mixed organophosphate and pyrethroid insecticide. Describe his/her condition and the extent of exposure.
It may be helpful to show this safety data sheet to physician.
As soon as a feeling of tingling is noted in any skin area (see section 11), it is recommended to immediately apply lidocaine or a vitamin E cream. For this purpose lidocaine or vitamin E cream should be available at the workplace.
In an industrial setting, the antidote atropine sulphate should be available at the workplace.
Malathion is a cholinesterase inhibitor affecting the central and peripheral nervous systems producing respiratory depression.
If allowed to penetrate the skin, gamma-cyhalothrin may cause an irritation similar to sunburn. The substance will be drawn into a non-polar environment such as a fat based oil or cream. Vitamin E cream has been reported to be beneficial.
Water is highly polar and will not decrease, but may prolong the irritation. Hot water may increase the pain.
For eye contamination, instillation of local anaesthetic can be considered.
For skin contamination, lidocaine can be used.
Decontamination procedures such as whole body washing, gastric lavage and administration of activated charcoal are often required.
Antidote: If symptoms of cholinesterase inhibition (see section 11) are present, administer atropine sulphate, which often is a

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lifesaving antidote, in large doses, TWO to FOUR mg intravenously or intramuscularly as soon as possible. Repeat at 5 to 10 minute intervals until signs of atropinisation appear and maintain full atropinisation until all organophosphate is metabolised.

Obidoxime chloride (Toxogonin), alternatively pralidoxime chloride(2-PAM), may be administered as an adjunct to, but not a substitute for atropine sulphate. Treatment with oxime should be maintained as long as atropine sulphate is administered.

At first sign of pulmonary oedema the patient should be given supplementary oxygen and treated symptomatically.

Relapse can occur after initial improvement. VERY CLOSE SUPERVISION OF THE PATIENT IS INDICATED FOR AT LEAST 48 HOURS, DEPENDING ON THE SEVERITY OF POISONING.

Much information on (acetyl)cholinesterase inhibition and its treatment can be found on the internet.

5. FIRE-FIGHTING MEASURES

- | | | |
|--|---|--|
| Suitable extinguishing media | : | Dry chemical, CO ₂ , water spray or regular foam.
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
| Unsuitable extinguishing media | : | Do not spread spilled material with high-pressure water streams.
High volume water jet |
| Specific hazards during fire fighting | : | Do not allow run-off from fire fighting to enter drains or water courses. |
| Hazardous combustion products | : | Fire may produce irritating, corrosive and/or toxic gases.
phosphorus oxides
Carbon oxides
Sulfur oxides
Nitrogen oxides (NO _x)
Fluorinated compounds
Chlorinated compounds
Hydrogen chloride
Hydrogen fluoride |
| Specific extinguishing methods | : | Remove undamaged containers from fire area if it is safe to do so.
Use a water spray to cool fully closed containers.
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. |
| Special protective equipment for fire-fighters | : | Wear self-contained breathing apparatus for firefighting if necessary. |

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6. ACCIDENTAL RELEASE MEASURES

- Personal precautions, protective equipment and emergency procedures : Use personal protective equipment.
Ensure adequate ventilation.
If it can be safely done, stop the leak.
Do not touch or walk through the spilled material.
Keep people away from and upwind of spill/leak.
Immediately evacuate personnel to safe areas.
Never return spills in original containers for re-use.
Mark the contaminated area with signs and prevent access to unauthorized personnel.
Only qualified personnel equipped with suitable protective equipment may intervene.
- Environmental precautions : Prevent product from entering drains.
Prevent further leakage or spillage if safe to do so.
If the product contaminates rivers and lakes or drains inform respective authorities.
- Methods and materials for containment and cleaning up : Never return spills in original containers for re-use.
Collect as much of the spill as possible with a suitable absorbent material.
Pick up and transfer to properly labeled containers.
Keep in suitable, closed containers for disposal.

7. HANDLING AND STORAGE

- Advice on protection against fire and explosion : Normal measures for preventive fire protection.
- Advice on safe handling : Do not breathe vapors/dust.
Avoid exposure - obtain special instructions before use.
Avoid contact with skin and eyes.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
To avoid spills during handling keep bottle on a metal tray.
Dispose of rinse water in accordance with local and national regulations.
Persons susceptible to skin sensitization problems or asthma, allergies, chronic or recurrent respiratory disease should not be employed in any process in which this mixture is being used.
- Conditions for safe storage : Prevent unauthorized access.
Keep container tightly closed in a dry and well-ventilated place.
Containers which are opened must be carefully resealed and kept upright to prevent leakage.
Observe label precautions.
Electrical installations / working materials must comply with the technological safety standards.
- Further information on storage : Protect from frost and extreme heat.

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age conditions

Store in closed, labelled containers. The storage room should be constructed of incombustible material, closed, dry, ventilated and with impermeable floor, without access of unauthorised persons or children. A warning sign reading "POISON" is recommended. The room should only be used for storage of chemicals. Food, drink, feed and seed should not be present. A hand wash station should be available.

The product is stable when stored at temperatures not exceeding 20 - 25°C.

The product should never be heated above 55°C. Local heating above this temperature should be avoided as well.

Recommended storage temperature : 5 - 25 °C

Further information on storage stability : No decomposition if stored and applied as directed.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Personal protective equipment

Respiratory protection : In the case of dust or aerosol formation use respirator with an approved filter.

Hand protection
Material : Wear chemical resistant gloves, such as barrier laminate, butyl rubber or nitrile rubber.

Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.

Eye protection : Eye wash bottle with pure water
Tightly fitting safety goggles
Wear face-shield and protective suit for abnormal processing problems.

Skin and body protection : Impervious clothing
Choose body protection according to the amount and concentration of the dangerous substance at the work place.

Protective measures : Plan first aid action before beginning work with this product. Always have on hand a first-aid kit, together with proper instructions.
Wear suitable protective equipment.
When using do not eat, drink or smoke.
In the context of professional plant protection use as recommended, the end user must refer to the label and the instructions for use.

Hygiene measures : Avoid contact with skin, eyes and clothing.
Do not inhale aerosol.

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When using do not eat or drink.
When using do not smoke.
Wash hands before breaks and at the end of workday.
Remove and wash contaminated clothing and gloves, including the inside, before re-use.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	: liquid
Color	: yellow
Odor	: Aromatic
Odor Threshold	: No data available
pH	: 2.49 (1% solution in water)
Melting point/freezing point	: No data available
Initial boiling point and boiling range	: No data available
Flash point	: 130 °C
Upper explosion limit / Upper flammability limit	: No data available
Lower explosion limit / Lower flammability limit	: No data available
Vapor pressure	: No data available
Relative vapor density	: No data available
Relative density	: 1.178 (20 °C)
Density	: 1.18 g/cm ³
Solubility(ies)	
Water solubility	: dispersible
Solubility in other solvents	: No data available
Partition coefficient: n-octanol/water	: No data available

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Autoignition temperature	:	No data available
Decomposition temperature	:	No data available
Viscosity		
Viscosity, dynamic	:	48 mPa.s (20 °C) 18 mPa.s (40 °C)
Viscosity, kinematic	:	No data available
Explosive properties	:	Not explosive
Oxidizing properties	:	The substance or mixture is not classified as oxidizing.
Particle size	:	No data available

10. STABILITY AND REACTIVITY

Reactivity	:	To our knowledge, the product has no special reactivities.
Chemical stability	:	Malathion will decompose rapidly when heated to temperatures above 140°C, significantly increasing the risk of explosion. Direct local heating such as electric heating or by steam must be avoided. The decomposition is dependent on time as well as temperature due to self-accelerating exothermic and autocatalytic reactions. The reactions involve rearrangements and polymerisation releasing volatile malodorous and inflammable compounds such as dimethyl sulphide and methyl mercaptan.
Possibility of hazardous reactions	:	None known
Conditions to avoid	:	Heat, flames and sparks. Heating of the product will produce harmful and irritant vapours. Protect from frost, heat and sunlight.
Incompatible materials	:	Strong alkalis, amines and strong oxidising compounds. The product can corrode metals (but does not meet the criteria for classification).

11. TOXICOLOGICAL INFORMATION

Acute toxicity

Toxic if swallowed.
Harmful in contact with skin or if inhaled.

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Product:

Acute oral toxicity	: LD50 Oral (Rat, female): 55 mg/kg Method: OECD Test Guideline 425 Symptoms: ataxia, hypoactivity, Breathing difficulties GLP: yes
Acute inhalation toxicity	: LC50 (Rat, male and female): > 2.07 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: OECD Test Guideline 403 Symptoms: hypoactivity, Breathing difficulties, ataxia GLP: yes
Acute dermal toxicity	: LD50 Dermal (Rat, male and female): > 5,000 mg/kg Method: OECD Test Guideline 402 GLP: yes

Components:

malathion (ISO) [containing ≤ 0,03 % isomalathion]:

Acute oral toxicity	: LD50 (Rat): 1,857 mg/kg Method: OECD Test Guideline 401 LD50 (Rat, female): 1,608 - 2,550 mg/kg Method: OECD Test Guideline 401 Symptoms: Tremors, hypoactivity GLP: yes
Acute inhalation toxicity	: LC50 (Rat): > 5.2 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: EPA OPP 81 - 3 GLP: yes Remarks: no mortality
Acute dermal toxicity	: LD50 (Rat): > 2,000 mg/kg Method: US EPA Test Guideline OPP 81-2 GLP: yes Assessment: The component/mixture is minimally toxic after single contact with skin. LD50 (Rat): > 2,000 mg/kg Method: OECD Test Guideline 402 GLP: yes Assessment: The component/mixture is minimally toxic after single contact with skin.

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Acute oral toxicity	: LD50 (Rat, male and female): > 5,000 mg/kg Method: OECD Test Guideline 401 Remarks: Based on data from similar materials
Acute inhalation toxicity	: LC50 (Rat): > 4.688 mg/l Exposure time: 4 h

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Test atmosphere: vapor
Assessment: The substance or mixture has no acute inhalation toxicity

Acute dermal toxicity : LD50 (Rabbit): > 2,000 mg/kg
Method: OECD Test Guideline 402
Assessment: The substance or mixture has no acute dermal toxicity

maleic anhydride:

Acute oral toxicity : LD50 (Rat, male and female): 1,090 mg/kg
Method: OECD Test Guideline 401

Acute dermal toxicity : LD50 (Rabbit, female): 2,620 mg/kg

GAMMA-CYHALOTHRIN:

Acute oral toxicity : LD50 (Rat, female): ca. 55 mg/kg
Method: OECD Test Guideline 401
Symptoms: Tremors
GLP: yes

LD50 (Rat, male): > 50 mg/kg
Method: OECD Test Guideline 401
Symptoms: Tremors
GLP: yes

Acute inhalation toxicity : LC50 (Rat, female): 0.0282 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
Symptoms: Tremors
GLP: yes

LC50 (Rat, male): 0.0402 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
Symptoms: Tremors
GLP: yes

Acute dermal toxicity : LD50 (Rat, female): 1,650 mg/kg
Method: OECD Test Guideline 402
Symptoms: Tremors
GLP: yes

LD50 (Rat, male): > 1,500 mg/kg
Method: OECD Test Guideline 402
Symptoms: Tremors
GLP: yes

phosphoric acid:

Acute oral toxicity : LD50 (Rat, female): 2,600 mg/kg
Method: OECD Test Guideline 423

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Skin corrosion/irritation

Causes skin irritation.

Product:

Species	: Rabbit
Method	: OECD Test Guideline 404
Result	: Severe skin irritation
GLP	: yes

Components:**malathion (ISO) [containing $\leq 0,03$ % isomalathion]:**

Species	: Rabbit
Method	: US EPA Test Guideline OPP 81-5
Result	: No skin irritation
GLP	: yes

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Species	: Rabbit
Assessment	: Repeated exposure may cause skin dryness or cracking.
Result	: No skin irritation
Remarks	: Minimal effects that do not meet the threshold for classification. Based on data from similar materials

Poly(oxy-1,2-ethanediyl), α -hydro- ω -hydroxy-, mono-C8-10-alkyl ethers, phosphates:

Method	: in vitro skin corrosion test
Result	: Corrosive after 3 minutes to 1 hour of exposure

maleic anhydride:

Species	: Rabbit
Exposure time	: 4 h
Result	: Corrosive after 3 minutes to 1 hour of exposure

GAMMA-CYHALOTHRIN:

Species	: Rabbit
Assessment	: Irritating to skin.
Method	: OECD Test Guideline 404
Result	: irritating
GLP	: yes

phosphoric acid:

Species	: Rabbit
Assessment	: Corrosive
Result	: Corrosive after 3 minutes to 1 hour of exposure

Serious eye damage/eye irritation

Causes serious eye damage.

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Product:

Species	: Rabbit
Assessment	: Risk of serious damage to eyes.
Method	: OECD Test Guideline 405
GLP	: yes
Remarks	: May cause irreversible eye damage.

Components:**malathion (ISO) [containing $\leq 0,03$ % isomalathion]:**

Species	: Rabbit
Result	: No eye irritation
Method	: EPA OPP 81-4
GLP	: yes

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Species	: Rabbit
Assessment	: No eye irritation
Remarks	: Minimal effects that do not meet the threshold for classification. Based on data from similar materials

Poly(oxy-1,2-ethanediyl), α -hydro- ω -hydroxy-, mono-C8-10-alkyl ethers, phosphates:

Result	: Irreversible effects on the eye
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maleic anhydride:

Species	: Rabbit
Result	: Irreversible effects on the eye

GAMMA-CYHALOTHRIN:

Species	: Rabbit
Result	: Eye irritation
Assessment	: Irritating to eyes.
Method	: OECD Test Guideline 405

phosphoric acid:

Result	: Irreversible effects on the eye
Remarks	: Based on skin corrosivity

Respiratory or skin sensitization**Skin sensitization**

May cause an allergic skin reaction.

Respiratory sensitization

Not classified based on available information.

Product:

Test Type	: Local lymph node assay (LLNA)
Routes of exposure	: Skin contact

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Assessment	:	May cause sensitization by skin contact.
Method	:	OECD Test Guideline 429
Result	:	Causes skin sensitization.
GLP	:	yes

Components:

malathion (ISO) [containing ≤ 0,03 % isomalathion]:

Routes of exposure	:	Dermal
Species	:	Guinea pig
Method	:	US EPA Test Guideline OPP 81-6
Result	:	Does not cause skin sensitization.
GLP	:	yes

Test Type	:	Local lymph node assay (LLNA)
Routes of exposure	:	Dermal
Species	:	mice
Method	:	OECD Test Guideline 429
Result	:	Does not cause skin sensitization.
GLP	:	yes

Test Type	:	Maximization Test
Routes of exposure	:	Dermal
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	May cause sensitization by skin contact.

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Test Type	:	Maximization Test
Species	:	Guinea pig
Result	:	Not a skin sensitizer.
Remarks	:	Based on data from similar materials

maleic anhydride:

Test Type	:	Local lymph node assay (LLNA)
Routes of exposure	:	Dermal
Species	:	Mouse
Assessment	:	The product is a skin sensitizer, sub-category 1A.
Method	:	OECD Test Guideline 429

GAMMA-CYHALOTHRIN:

Routes of exposure	:	Skin contact
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	May cause sensitization by skin contact.

Germ cell mutagenicity

Not classified based on available information.

Components:

malathion (ISO) [containing ≤ 0,03 % isomalathion]:

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Genotoxicity in vitro : Test Type: Ames test
Result: negative

Test Type: In vitro mammalian cell gene mutation test
Result: positive
Remarks: Based on data from similar materials

Test Type: unscheduled DNA synthesis assay
Result: negative
Remarks: Based on data from similar materials

Genotoxicity in vivo : Test Type: chromosome aberration assay
Species: Rat
Result: negative
Remarks: Based on data from similar materials

Test Type: unscheduled DNA synthesis assay
Species: Rat
Result: negative
Remarks: Based on data from similar materials

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Genotoxicity in vitro : Test Type: reverse mutation assay
Method: OECD Test Guideline 471
Result: negative
Remarks: Based on data from similar materials

Genotoxicity in vivo : Test Type: Bone marrow chromosome aberration.
Species: Rat
Application Route: inhalation (vapor)
Result: negative

maleic anhydride:

Genotoxicity in vitro : Test Type: reverse mutation assay
Method: OECD Test Guideline 471
Result: negative

Test Type: In vitro mammalian cell gene mutation test
Method: OECD Test Guideline 476
Result: negative
Remarks: Based on data from similar materials

Genotoxicity in vivo : Test Type: Bone marrow chromosome aberration.
Species: Rat (male and female)
Application Route: Inhalation
Method: OECD Test Guideline 475
Result: negative

Germ cell mutagenicity - Assessment : Weight of evidence does not support classification as a germ cell mutagen.

GAMMA-CYHALOTHRIN:

Genotoxicity in vitro : Test Type: Ames test

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Metabolic activation: with and without metabolic activation
Result: negative

Method: OECD Test Guideline 476
Result: negative

Genotoxicity in vivo : Test Type: Micronucleus test
Species: Mouse
Result: negative
GLP: yes

phosphoric acid:

Genotoxicity in vitro : Test Type: reverse mutation assay
Method: OECD Test Guideline 471
Result: negative

Test Type: Chromosome aberration test in vitro
Method: OECD Test Guideline 473
Result: negative

Carcinogenicity

Not classified based on available information.

Components:

malathion (ISO) [containing $\leq 0,03$ % isomalathion]:

Species : Rat
Application Route : Ingestion
Exposure time : 24 month(s)
NOAEL : 6,000 ppm
Result : positive

Remarks : Probably carcinogenic to humans (IARC 2A)

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Species : Rat, male and female
Application Route : inhalation (vapor)
Exposure time : 12 month(s)
NOAEC : 1.8 mg/l
Result : negative
Remarks : Based on data from similar materials

Carcinogenicity - Assessment : Not classifiable as a human carcinogen.

maleic anhydride:

Species : Rat, male and female
Application Route : Oral
Exposure time : 2 Years
Dose : 0, 10, 32, 100 mg/kg body weight
NOEL : 10 mg/kg body weight
Method : OECD Test Guideline 451

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Result : negative

Carcinogenicity - Assessment : Weight of evidence does not support classification as a carcinogen

Reproductive toxicity

Not classified based on available information.

Components:

malathion (ISO) [containing ≤ 0,03 % isomalathion]:

Effects on fertility : Test Type: Two-generation study
Species: Rat, male and female
General Toxicity F1: NOAEL: 132 - 152 mg/kg bw/day
Symptoms: Reduced offspring weight gain.

Effects on fetal development : Test Type: Embryo-fetal development
Species: Rat
General Toxicity Maternal: NOAEL: 400 mg/kg bw/day
Teratogenicity: NOAEL: 800 mg/kg bw/day
Result: No teratogenic effects.

Test Type: Embryo-fetal development
Species: Rabbit
General Toxicity Maternal: NOAEL: 25 mg/kg bw/day
Teratogenicity: NOAEL: 25 mg/kg bw/day
Result: No teratogenic effects.

maleic anhydride:

Effects on fertility : Test Type: Two-generation study
Species: Rat, male and female
Application Route: Oral
Dose: 0, 20, 55, and 150 milligram per kilogram
General Toxicity Parent: LOAEL: 20 mg/kg body weight
Fertility: NOEL: 55 mg/kg body weight
Method: OECD Test Guideline 416
Result: negative

Effects on fetal development : Species: Rat
Application Route: Oral
Duration of Single Treatment: 15 d
General Toxicity Maternal: NOAEL: ≥ 140 mg/kg body weight
Teratogenicity: NOAEL: ≥ 140 mg/kg body weight
Embryo-fetal toxicity.: NOAEL: ≥ 140 mg/kg body weight
Method: OECD Test Guideline 414
Result: negative

Reproductive toxicity - Assessment : Weight of evidence does not support classification for reproductive toxicity

GAMMA-CYHALOTHRIN:

Effects on fetal development : Species: Rat
Dose: 1, 2.5, 5, 10 or 15 mg/kg bw/day

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Embryo-fetal toxicity.: NOEL: 2.5 mg/kg bw/day

phosphoric acid:

Effects on fertility : Test Type: reproductive and developmental toxicity study
Species: Rat, male and female
Application Route: Ingestion
General Toxicity Parent: NOAEL: 500 mg/kg body weight
General Toxicity F1: NOAEL: 500 mg/kg body weight
Method: OECD Test Guideline 422
Result: negative

Effects on fetal development : Test Type: Embryo-fetal development
Species: Mouse
Application Route: Ingestion
General Toxicity Maternal: NOAEL: 370 mg/kg body weight
Developmental Toxicity: NOAEL: 370 mg/kg body weight
Result: negative
Remarks: Based on data from similar materials

STOT-single exposure

Not classified based on available information.

STOT-repeated exposure

May cause damage to organs (Nervous system) through prolonged or repeated exposure.
May cause damage to organs (Respiratory system) through prolonged or repeated exposure if inhaled.

Components:

maleic anhydride:

Routes of exposure : inhalation (dust/mist/fume)
Target Organs : Respiratory system
Assessment : The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 1.

GAMMA-CYHALOTHRIN:

Target Organs : Nervous system
Assessment : The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 1.

Repeated dose toxicity

Components:

malathion (ISO) [containing ≤ 0,03 % isomalathion]:

Species : Rat
LOAEL : 34.4 mg/kg
Application Route : Oral - feed
Exposure time : 90 d
Target Organs : Nervous system
Symptoms : cholinesterase inhibition

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

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Species	: Rat, male and female
NOAEC	: 0.9 - 1.8 mg/l
Application Route	: inhalation (vapor)
Exposure time	: 12 Months

maleic anhydride:

Species	: Dog, male and female
NOAEL	: 60 mg/kg
Application Route	: Oral
Exposure time	: 90 d
Dose	: 0, 20, 40, or 60 mg/kg bw/day
Method	: OECD Test Guideline 409

Species	: Rat, male and female
NOEL	: 10 mg/kg
Application Route	: Oral
Exposure time	: 2 years
Dose	: 0, 10, 32, and 100 mg/kg bw/day
Method	: OECD Test Guideline 452

Species	: Rat, male and female
LOAEC	: 0.0011 mg/l
Application Route	: Inhalation
Exposure time	: 6 months
Target Organs	: Respiratory system

GAMMA-CYHALOTHRIN:

Species	: Rat, male and female
NOAEL	: 50 ppm
Application Route	: Oral - feed
Exposure time	: 13 weeks

Species	: Rat, male and female
NOAEL	: 4.19 - 4.49 mg/kg
LOAEL	: 8.81 - 10.24 mg/kg
Application Route	: Oral - feed
Exposure time	: 13 weeks
Method	: OECD Test Guideline 407
Target Organs	: Nervous system
Symptoms	: decrease in appetite

phosphoric acid:

Species	: Rat, male and female
NOAEL	: 250 mg/kg
Application Route	: Oral - gavage
Exposure time	: 42 - 54 d
Method	: OECD Test Guideline 422

Aspiration toxicity

May be fatal if swallowed and enters airways.

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Components:**malathion (ISO) [containing $\leq 0,03$ % isomalathion]:**

The substance does not have properties associated with aspiration hazard potential.

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

May be fatal if swallowed and enters airways.

GAMMA-CYHALOTHRIN:

The substance does not have properties associated with aspiration hazard potential.

Experience with human exposure**Components:****Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:**

Skin contact : Symptoms: Repeated exposure may cause skin dryness or cracking.

Further information**Product:**

Remarks : Solvents may degrease the skin.
On contact, the first symptoms to appear may be irritation and allergic reactions. Symptoms of cholinesterase inhibition: nausea, headache, vomiting, cramps, weakness, blurred vision, pin-point pupils, tightness in chest, laboured breathing, nervousness, sweating, watering of eyes, drooling or frothing of mouth and nose, muscle spasms and coma.

Components:**malathion (ISO) [containing $\leq 0,03$ % isomalathion]:**

Remarks : The active ingredient malathion is a cholinesterase inhibitor of low mammalian toxicity. However, prolonged storage or storage at too high temperatures may induce formation of the much more toxic and synergistic contaminant isomalathion (LD50, oral, rat, 89 mg/kg). Both malathion and isomalathion rapidly enter the body on contact with all skin surfaces and eyes.
Repeated exposures to cholinesterase inhibitors such as malathion or isomalathion may, without warning, cause increased susceptibility to doses of any cholinesterase inhibitor.

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Remarks : Vapour concentrations above recommended exposure levels are irritating to the eyes and the respiratory tract, may cause headaches and dizziness, are anaesthetic and may have other central nervous system effects. Prolonged and/or repeated skin contact with low viscosity materials may defat the skin resulting in possible irritation and dermatitis. Small amounts of

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liquid aspirated into the lungs during ingestion or from vomiting may cause chemical pneumonitis or pulmonary edema.

12. ECOLOGICAL INFORMATION

Ecotoxicity

Product:

Toxicity to daphnia and other aquatic invertebrates	:	LC50 (<i>Daphnia magna</i> (Water flea)): 1.99 µg/l Exposure time: 48 h
Toxicity to soil dwelling organisms	:	LC50 (<i>Eisenia fetida</i> (earthworms)): 129 mg/kg Exposure time: 14 d
Toxicity to terrestrial organisms	:	LD50 (<i>Coturnix japonica</i> (Japanese quail)): 215 mg/kg
		LC50 (<i>Apis mellifera</i> (bees)): 0.19 Exposure time: 48 h

Components:

malathion (ISO) [containing ≤ 0,03 % isomalathion]:

Toxicity to fish	:	LC50 (<i>Oncorhynchus mykiss</i> (rainbow trout)): 0.18 mg/l Exposure time: 96 h
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (<i>Daphnia similis</i> (Water flea)): 1.71 µg/l Exposure time: 48 h
Toxicity to algae/aquatic plants	:	IC50 (<i>Selenastrum capricornutum</i> (green algae)): 4.06 mg/l Exposure time: 72 h
M-Factor (Acute aquatic toxicity)	:	100
Toxicity to fish (Chronic toxicity)	:	NOEC (<i>Oncorhynchus mykiss</i> (rainbow trout)): 0.021 mg/l Exposure time: 37 d
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC (<i>Daphnia magna</i> (Water flea)): 0.00006 mg/l Exposure time: 21 d
M-Factor (Chronic aquatic toxicity)	:	1,000
Toxicity to soil dwelling organisms	:	(<i>Eisenia fetida</i> (earthworms)): 613 mg/kg Exposure time: 14 d

Remarks: No significant adverse effect on Nitrogen mineralization.
No significant adverse effect on Carbon mineralization.

Toxicity to terrestrial organisms	:	LD50 (<i>Colinus virginianus</i> (Bobwhite quail)): 359 mg/kg
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Exposure time: 5 d

LC50 (Colinus virginianus (Bobwhite quail)): 3,497 mg/kg
Exposure time: 5 d
Remarks: Dietary

LD50 (Anas platyrhynchos (Mallard duck)): > 2,250 mg/kg

LD50 (Apis mellifera (bees)): 0.38 µg/bee
End point: Acute oral toxicity

Ecotoxicology Assessment

Toxicity Data on Soil : Harmful to the soil environment.

Other organisms relevant to the environment : Harmful to terrestrial vertebrates., Harmful to terrestrial invertebrates.

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Toxicity to fish : LL50 (Oncorhynchus mykiss (rainbow trout)): 2 - 5 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EL50 (Daphnia magna (Water flea)): 1.4 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : EL50 (Pseudokirchneriella subcapitata (green algae)): 1 - 3 mg/l
Exposure time: 24 h
Method: OECD Test Guideline 201

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : EL50 (Daphnia magna (Water flea)): 0.89 mg/l
Exposure time: 21 d
Method: OECD Test Guideline 211

Toxicity to microorganisms : LL50 (Tetrahymena pyriformis): 677.9 mg/l
Exposure time: 72 h
Test Type: Growth inhibition

Poly(oxy-1,2-ethanediyl), α-hydro-ω-hydroxy-, mono-C8-10-alkyl ethers, phosphates:

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 8.8 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : NOEC (Desmodesmus subspicatus (green algae)): 6.25 mg/l
Exposure time: 72 h
Test Type: semi-static test

ErC50 (Desmodesmus subspicatus (green algae)): 63 - 78 mg/l
Exposure time: 72 h
Test Type: semi-static test

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maleic anhydride:

Toxicity to daphnia and other aquatic invertebrates : EC50 (*Daphnia magna* (Water flea)): 42.81 mg/l
 Exposure time: 48 h
 Method: OECD Test Guideline 202
 Remarks: Based on data from similar materials

Toxicity to algae/aquatic plants : EC10 (*Pseudokirchneriella subcapitata* (green algae)): 11.8 mg/l
 Exposure time: 72 h
 Method: OECD Test Guideline 201
 Remarks: Based on data from similar materials

EC50 (*Pseudokirchneriella subcapitata* (green algae)): 74.35 mg/l
 Exposure time: 72 h
 Method: OECD Test Guideline 201
 Remarks: Based on data from similar materials

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC (*Daphnia magna* (Water flea)): 10 mg/l
 Exposure time: 21 d

Toxicity to microorganisms : EC10 (*Pseudomonas putida*): 44.6 mg/l
 Exposure time: 18 h
 Method: DIN 38 412 Part 8

GAMMA-CYHALOTHRIN:

Toxicity to fish : LC50 (*Oncorhynchus mykiss* (rainbow trout)): 0.07 µg/l
 Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (*Daphnia magna* (Water flea)): 0.1 µg/l
 Exposure time: 48 h
 Test Type: Static renewal test
 Method: OECD Test Guideline 202

(*Hyalella azteca* (Amphipod)): 0.000086 µg/l
 Exposure time: 96 h
 Test Type: flow-through test
 Method: OPPTS 850.1010

Toxicity to algae/aquatic plants : EC50 (algae): > 2.85 mg/l
 Exposure time: 72 h

NOEC (*Lemna gibba* (duckweed)): 0.5 µg/l
 Exposure time: 7 d
 Method: OECD Test Guideline 221

M-Factor (Acute aquatic toxicity) : 10,000

Toxicity to fish (Chronic toxicity) : NOEC (*Pimephales promelas* (fathead minnow)): 0.016 µg/l
 End point: mortality
 Exposure time: 7 d
 Test Type: Early Life-Stage

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GLP: yes

LOEC (Pimephales promelas (fathead minnow)): 0.04 µg/l
End point: mortality
Exposure time: 7 d
Test Type: Early Life-Stage
GLP: yes

NOEC (Pimephales promelas (fathead minnow)): 0.0379 µg/l
End point: Hatching success
Exposure time: 35 d
Test Type: flow-through test
GLP: yes

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC (Daphnia magna (Water flea)): 0.0019 µg/l
End point: reproduction
Exposure time: 21 d
Test Type: flow-through test
Method: OECD Test Guideline 211

M-Factor (Chronic aquatic toxicity) : 10,000

Toxicity to soil dwelling organisms : LC50 (Eisenia fetida (earthworms)): > 1300 mg/kg dry weight (d.w.)
Exposure time: 14 d

Toxicity to terrestrial organisms : LD50 (Colinus virginianus (Bobwhite quail)): > 2,000 mg/kg

LD50 (Apis mellifera (bees)): 0.005 µg/bee
Exposure time: 24 h
End point: Acute contact toxicity

LD50 (Apis mellifera (bees)): 4.2 µg/bee
Exposure time: 24 h
End point: Acute oral toxicity

phosphoric acid:

Toxicity to fish : LC50 (Lepomis macrochirus (Bluegill sunfish)): 3 - 3.25 mg/l
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): > 100 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : EC50 (Desmodesmus subspicatus (green algae)): > 100 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

NOEC (Desmodesmus subspicatus (green algae)): 100 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

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Toxicity to microorganisms : EC50 (activated sludge): > 1,000 mg/l
Exposure time: 3 h
Method: OECD Test Guideline 209

Persistence and degradability**Product:**

Biodegradability : Remarks: Product contains minor amounts of not readily biodegradable components, which may not be degradable in waste water treatment plants.

Components:**malathion (ISO) [containing ≤ 0,03 % isomalathion]:**

Biodegradability : Result: Not readily biodegradable.

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Biodegradability : Result: Readily biodegradable.
Biodegradation: 58.6 %
Exposure time: 28 d
Method: OECD Test Guideline 301F
Remarks: Based on data from similar materials

Poly(oxy-1,2-ethanediyl), α-hydro-ω-hydroxy-, mono-C8-10-alkyl ethers, phosphates:

Biodegradability : Result: Biodegradable
Biodegradation: 87 %
Exposure time: 28 d
Method: Regulation (EC) No. 440/2008, Annex, C.4-B

maleic anhydride:

Biodegradability : Inoculum: activated sludge, non-adapted
Result: Readily biodegradable.
Biodegradation: > 90 %
Exposure time: 25 d
Method: OECD Test Guideline 301B
Remarks: Based on data from similar materials

GAMMA-CYHALOTHRIN:

Biodegradability : Result: Not readily biodegradable.
Biodegradation: 21 %
Exposure time: 28 d

phosphoric acid:

Biodegradability : Remarks: The methods for determining biodegradability are not applicable to inorganic substances.

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Bioaccumulative potential**Components:****malathion (ISO) [containing ≤ 0,03 % isomalathion]:**

Bioaccumulation : Species: Fish
Bioconcentration factor (BCF): 95
Remarks: Bioaccumulation is unlikely.

Partition coefficient: n-octanol/water : log Pow: 2.75

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Bioaccumulation : Remarks: The product/substance has a potential to bioaccumulate.

Partition coefficient: n-octanol/water : log Pow: 3.72
Method: QSAR

maleic anhydride:

Bioaccumulation : Remarks: Bioaccumulation is unlikely.

Partition coefficient: n-octanol/water : log Pow: -2.61

GAMMA-CYHALOTHRIN:

Bioaccumulation : Remarks: Can accumulate in aquatic organisms.

Partition coefficient: n-octanol/water : log Pow: 4.96 (19 °C)
Method: OECD Test Guideline 107

log Pow: 5.65
Method: OECD Test Guideline 117

Mobility in soil**Components:****malathion (ISO) [containing ≤ 0,03 % isomalathion]:**

Distribution among environmental compartments : Remarks: medium mobility in soil

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Distribution among environmental compartments : Remarks: Expected to partition to sediment and wastewater solids. Moderately volatile.

GAMMA-CYHALOTHRIN:

Distribution among environmental compartments : Remarks: immobile

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Other adverse effects

Product:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Very toxic to aquatic life with long lasting effects.

Components:

GAMMA-CYHALOTHRIN:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Very toxic to aquatic life with long lasting effects.

phosphoric acid:

Additional ecological information : Harmful effects on aquatic organisms also due to pH shift.

Hygienic standards:

(Allowable concentration in air, water, including fishery waters, soil)

Components	Air	Water	Soil	Data Source
malathion (ISO) [containing ≤ 0,03 % iso-malathion] 121-75-5	MPC - maximum: 0.015 mg/m ³ Limiting health hazard indicator: reflectory Hazard class: Class 2 - highly dangerous	MPC: 0.00001 Milligrams per cubed decimeter Limiting health hazard indicator: toxic Hazard class: 1	No data available	List 1 List 5
Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified 64742-94-5	TSEL: 0.2 mg/m ³	MPC: 0.05 Milligrams per cubed decimeter Limiting health hazard indicator: toxic Hazard class: 3	No data available	List 2 List 5
maleic anhydride 108-31-6	MPC - average: 0.05 mg/m ³ Limiting health hazard indicator: Reflectory-resorptive Hazard class: Class 2 - highly dangerous MPC - maximum: 0.2 mg/m ³ Limiting health hazard indicator: Reflectory-resorptive Hazard class: Class	MPC: 0.01 Milligrams per cubed decimeter Limiting health hazard indicator: toxic Hazard class: 4	No data available	List 1 List 5

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	2 - highly dangerous			
phosphoric acid 7664-38-2	TSEL: 0.02 mg/m3	MPC: 0.01 Milligrams per cubed decimeter (in recalc on sub- stance 0.5) Limiting health hazard indicator: toxic Hazard class: 4e	No data avail- able	List 2 List 5

For explanation of abbreviations see section 16.

13. DISPOSAL CONSIDERATIONS

Disposal methods

- Waste from residues : The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Send to a licensed waste management company.
- Contaminated packaging : Empty remaining contents.
Triple rinse containers.
Do not re-use empty containers.
Packaging that is not properly emptied must be disposed of as the unused product.
Empty containers should be taken to an approved waste handling site for recycling or disposal.

14. TRANSPORT INFORMATION

ADR

- UN number : UN 2902
Proper shipping name : PESTICIDE, LIQUID, TOXIC, N.O.S.
(Malathion, Gamma-cyhalothrin, ALKYL(C3-C6)BENZENES)
Class : 6.1
Packing group : III
Labels : 6.1
Hazard Identification Number : 60
Tunnel restriction code : (E)
Environmentally hazardous : yes

UNRTDG

- UN number : UN 2902
Proper shipping name : PESTICIDE, LIQUID, TOXIC, N.O.S.
(Malathion, Gamma-cyhalothrin, ALKYL(C3-C6)BENZENES)
Class : 6.1
Packing group : III
Labels : 6.1

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Environmentally hazardous : yes

IATA-DGR

UN/ID No. : UN 2902
Proper shipping name : Pesticide, liquid, toxic, n.o.s.
(Malathion, Gamma-cyhalothrin, ALKYL(C3-C6)BENZENES)
Class : 6.1
Packing group : III
Labels : Toxic
Packing instruction (cargo aircraft) : 663
Packing instruction (passenger aircraft) : 655
Environmentally hazardous : yes

IMDG-Code

UN number : UN 2902
Proper shipping name : PESTICIDE, LIQUID, TOXIC, N.O.S.
(Malathion, Gamma-cyhalothrin, ALKYL(C3-C6)BENZENES)
Class : 6.1
Packing group : III
Labels : 6.1
EmS Code : F-A, S-A
Marine pollutant : yes

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

The ingredients of this product are reported in the following inventories:

TCSI : On the inventory, or in compliance with the inventory
TSCA : Product contains substance(s) not listed on TSCA inventory.
AIIC : Not in compliance with the inventory
DSL : This product contains the following components that are not on the Canadian DSL nor NDSL.
(S)- α -CYANO-3-PHENOXYBENZYL (1R,3R)-3-[(Z)-2-CHLORO-3,3,3-TRIFLUOROPROP-1-ENYL]-2,2-DIMETHYLCYCLOPROPANECARBOXYLATE
ENCS : Not in compliance with the inventory

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ISHL	:	Not in compliance with the inventory
KECI	:	Not in compliance with the inventory
PICCS	:	Not in compliance with the inventory
IECSC	:	Not in compliance with the inventory
NZIoC	:	Not in compliance with the inventory
TECI	:	Not in compliance with the inventory

16. OTHER INFORMATION

Full text of H-Statements

H290	May be corrosive to metals.
H301	Toxic if swallowed.
H302	Harmful if swallowed.
H303	May be harmful if swallowed.
H304	May be fatal if swallowed and enters airways.
H312	Harmful in contact with skin.
H313	May be harmful in contact with skin.
H314	Causes severe skin burns and eye damage.
H315	Causes skin irritation.
H317	May cause an allergic skin reaction.
H318	Causes serious eye damage.
H319	Causes serious eye irritation.
H330	Fatal if inhaled.
H333	May be harmful if inhaled.
H372	Causes damage to organs through prolonged or repeated exposure.
H372	Causes damage to organs through prolonged or repeated exposure if inhaled.
H400	Very toxic to aquatic life.
H401	Toxic to aquatic life.
H402	Harmful to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.
H411	Toxic to aquatic life with long lasting effects.

Full text of other abbreviations

Acute Tox.	:	Acute toxicity
Aquatic Acute	:	Short-term (acute) aquatic hazard
Aquatic Chronic	:	Long-term (chronic) aquatic hazard
Asp. Tox.	:	Aspiration hazard
Eye Dam.	:	Serious eye damage
Eye Irrit.	:	Eye irritation
Met. Corr.	:	Corrosive to Metals
Skin Corr.	:	Skin corrosion
Skin Irrit.	:	Skin irritation
Skin Sens.	:	Skin sensitization
STOT RE	:	Specific target organ toxicity - repeated exposure
List 1	:	SanPiN 1.2.3685-21 Table 1.1 Maximum permissible concentration (MPC) of pollutants in the air of urban and rural settle-

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- ments
- List 2 : SanPiN 1.2.3685-21 Table 1.2 Tentative Safe Exposure Levels (TSEL) of pollutants in the air of urban and rural settlements
- List 5 : Order of the Russian Federal Fisheries Agency "Standards of maximum permissible concentrations of harmful substances in fishery water bodies"

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - substance of very high concern; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TRGS - Technical Rule for Hazardous Substances; TSCA - Toxic Substances Control Act (United States); UN - United Nations; vPvB - Very Persistent and Very Bioaccumulative

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SAFETY DATA SHEET



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Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	01.09.2025	50001281	Date of first issue: 01.09.2025

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